



Rehnuma (رهنما)

Project Proposal & System Design

Faizan Shaikh, 2 September 2025.

Rehnuma (رهنما) :

Your Personal AI Study Partner.

- Turn Your Notes Into a Conversation.

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The Silent Struggle of Every Student

Globally, students are working harder than ever, but their results don't always reflect their effort. The root of the problem isn't a lack of material; it's the outdated, inefficient way we study.

- **Information Overload:** Students are drowning in a sea of digital notes, PDFs, and textbooks.
- **Time Wasted:** Countless hours are lost simply searching for specific facts and formulas before exams.
- **High Stress & Burnout:** The pressure of university exams, standardized tests (like the SAT or GRE), and professional certifications is immense.

The Real Enemy : Passive Learning

The traditional method of re-reading notes is scientifically proven to be ineffective for long-term memory.

- **The Forgetting Curve:** We forget most of what we learn within days if we don't actively recall it.
- **False Confidence:** Passive reading creates an "illusion of knowing" that is shattered under exam pressure.
- **Lack of Engagement:** It's a one-way, boring process that fails to engage the mind.

The Opportunity

A Global, Growing Market

The demand for smarter learning tools is a massive global trend.

- **250 Million+** students in higher education worldwide.
- A global EdTech market projected to exceed **\$400 billion** by 2027.
- A generation of digital-native learners stuck with analog study habits.

This is a universal problem in search of a modern solution.

The Solution

Introducing Rehnuma: Your Guide to Smarter Studying

Rehnuma is an intelligent web application that transforms any study material into a dynamic, interactive learning partner.

It's not just a PDF reader; it's a conversation with your notes.

The Core Experience

Two Modes, One Powerful Goal: Active Learning

Rehnuma brings notes to life through two distinct, powerful modes.

- **Exploration Mode (Search & Understand):**

- Instantly find any information by asking questions in plain English.

- **Challenge Mode (Test & Retain):**

- Let the AI quiz you on your own notes to build deep, lasting knowledge.

Demo Flow - 1

Exploration Mode : Your Personal Search Engine

1. **Upload:** A student uploads their "World History" PDF notes.
2. **Ask:** In the chat interface, they type, *"What were the main consequences of the Treaty of Versailles?"*
3. **Get Answer:** Instantly, Rehnuma provides a concise summary, sourced directly from their notes, with references to the exact page or section.

Outcome: Doubt cleared in 10 seconds, not 10 minutes.

Demo Flow - 2

Challenge Mode : The Ultimate Revision Tool

1. **Select:** The student chooses their "Organic Chemistry" notes and starts a quiz.
2. **AI Generates:** Rehnuma creates a question: *"Explain the mechanism of an SN2 reaction."*
3. **Answer & Feedback:** The student types their answer. Rehnuma verifies it and provides instant feedback: *"Correct! You mentioned the single-step mechanism and inversion of configuration."*
4. **Review:** If the answer was wrong, Rehnuma shows the ideal answer and a button: **"Show me in my notes,"** highlighting the exact paragraph for review.

Outcome: Active recall solidifies memory and identifies weaknesses.

The Technology Behind Magic

Retrieval - Augmented Generation (RAG)

Rehnuma is powered by a state-of-the-art AI pipeline that runs privately and securely.

1.Index: When a document is uploaded, it's broken down and its "meaning" is stored in a vector database.

2.Retrieve: When you ask a question, the AI finds the most relevant pieces of information from your notes.

3.Generate: A local Large Language Model (LLM) uses this retrieved information to generate a coherent, accurate answer or a relevant quiz question.

Key Benefit: All AI processing is done with open-source models. Your notes are never sent to third-party APIs.

System Architecture

Built For Scale, Security, and Speed

Our decoupled architecture ensures a smooth user experience, even during intensive AI processing.

Tech Stack Summary

Modern, Robust & Open-Source

- **Frontend:** Next.js (React), TypeScript, Tailwind CSS
- **Backend API:** Python, FastAPI
- **AI Worker:** Celery with Redis
- **AI & ML:** LangChain, Sentence-Transformers, Ollama (Llama 3 8B)
- **Databases:** PostgreSQL, ChromaDB
- **Deployment:** Docker, AWS / DigitalOcean

Competitive Landscape

Where Rehnuma Fits

Rehnuma operates in a unique space, combining active learning with hyper-personalization.

- **Generic & Passive (Bottom-Left):** Standard PDF Readers (Adobe), Note-taking apps (Evernote).
- **Generic & Active (Bottom-Right):** Generic flashcard apps (Quizlet, Anki).
- **Personalized & Passive (Top-Left):** *This quadrant is largely empty.*
- **Personalized & Active (Top-Right):** **Rehnuma.** We are the only tool that actively quizzes you on your own material.

Business Model

Freemium : Accesible to All, Valuable to Power Users

Free Tier (Rehnuma Basic):

- Upload up to 3 documents.
- Limited monthly questions & quizzes.
- **Goal:** Let every student experience the core magic.

Pro Tier (Rehnuma Pro):

- **Price:** \$5 USD/month (or a discounted \$50/year).
- Unlimited uploads, questions, and quizzes.
- Advanced features like folder organization.
- **Goal:** Serve the needs of serious students and dedicated learners.

The Roadmap

Journey : From MVP to a Learning Community

Phase 1 (MVP - Next 4 Months):

- Launch core product: PDF upload, Exploration Mode, and Challenge Mode.
- Onboard first 1,000 users.

Phase 2 (Expansion - 2026):

- Introduce folder organization and support for more file types (DOCX, PPTX).
- Develop user progress analytics.

Phase 3 (Community - Late 2026):

- Implement features for sharing notes and quizzes.
- Explore collaborative study groups.

Vision & The Ask

To make Learning Personal

Our vision is to empower every student, everywhere, to study smarter, not just harder. We believe personalized, active learning is the key to unlocking their true potential.



“Thank You”

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